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import java.io.*;
import java.util.*;

class Account implements Serializable {
    private static final long serialVersionUID = 1L;
    String owner;
    int balance;
    List<String> history = new ArrayList<>();

    Account(String owner){
        this.owner = (owner == null || owner.isBlank()) ? "Unknown" : owner.trim();
        history.add("Account created");
    }

    void deposit(int amount){
        balance += amount;
        history.add("Deposit: " + amount + " (bal=" + balance + ")");
    }

    boolean withdraw(int amount){
        if (amount > balance) return false;
        balance -= amount;
        history.add("Withdraw: " + amount + " (bal=" + balance + ")");
        return true;
    }
}

public class BankApp {
    private static final String DATA_FILE = "account.ser";
    private Account account;
    private final Scanner sc = new Scanner(System.in);

    public static void main(String[] args){
        BankApp app = new BankApp();
        app.load();
        app.run();
    }

    private void run(){
        if (account == null){
            System.out.print("Owner name: ");
            account = new Account(sc.nextLine());
        }
        for(;;){
            menu();
            switch(sc.nextLine().trim()){
                case "1" -> info();
                case "2" -> deposit();
                case "3" -> withdraw();
                case "0" -> { save(); System.out.println("Bye."); return; }
                default -> System.out.println("Unknown.");
            }
        }
    }

    private void menu(){
        System.out.println("\n\n== Mini Bank ==");
        System.out.println("1.Info 2.Deposit 3.Withdraw 0.Exit");
        System.out.print("> ");
    }

    private void info(){
        System.out.println("Owner: " + account.owner);
        System.out.println("Bal : " + account.balance);
    }

    private void deposit(){
        System.out.print("Deposit amount: ");
        int x = parsePositive(sc.nextLine());
        if (x <= 0) return;
        account.deposit(x);
        System.out.println("OK");
    }

    private void withdraw(){
        System.out.print("Withdraw amount: ");
        int x = parsePositive(sc.nextLine());
        if (x <= 0) return;
        System.out.println(account.withdraw(x) ? "OK" : "Not enough");
    }

    private int parsePositive(String s){
        try{
            int x = Integer.parseInt(s.trim());
            if (x <= 0) System.out.println("Must be positive.");
            return x;
        }catch(Exception e){
            System.out.println("Invalid number.");
            return -1;
        }
    }

    private void load(){
        File f = new File(DATA_FILE);
        if (!f.exists()) return;
        try(ObjectInputStream in = new ObjectInputStream(new FileInputStream(f))){
            account = (Account) in.readObject();
            System.out.println("Loaded: " + account.owner);
        }catch(Exception e){
            System.out.println("Load failed: " + e.getMessage());
        }
    }

    try(ObjectOutputStream out = new ObjectOutputStream(new FileOutputStream(DATA_FILE))){ out.writeObject(account); System.out.println("Saved.");
    }catch(IOException e){ System.out.println("Save failed: " + e.getMessage()); }
}

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